

Major Data Analysis Project Report

Group 2

Mazen Imad Jaber - n9940669

Aidan Lockwood

Nima Ghamari

Introduction

The aim of the project is to develop a robust credit risk model using GLMs and GLMMs to predict loan default probability for a recently acquired peer-to-peer lending start up by a regional Australian bank. The main concern for the bank is that all the previous credit risk models, which were employed by the start up, have been ad hoc without required rigors with a view to meeting strict regulatory standards.

Project Objectives

The primary objective is to rebuild the credit risk model from the ground up, employing advanced statistical techniques to improve its predictive accuracy. This involves addressing the following key questions:

How does the new model perform compared to the old model with respect to the defaults of loans? What are the critical driving variables of credit risk, and what are their relationships to the ones in traditional banking sectors? Can accounting for variations over time and geography - state or zip code, say - improve this model's performance? In order to meet these needs, we used two models: the Generalized Linear Model (GLM): For the first two questions to optimise model performance and get the important variables; the Generalized Linear Mixed Effects Model (GLMM): Discriminating between time and geographical variation as suggested by the industry expert for question(s) concerning jurisdictional differences in credit risk.

Key Analysis Focus

Our objective in modeling will be to derive models that predict loan defaulters using information available to the lender at the time the loan is applied for. Using Random Forests and selection of either the most relevant predictors, or Stepwise Selection via the Akaike information criterion, variable selection is made. Its performance should also be benchmarked on Gini scores, ROC curves, and other benchmark industry metrics to ensure the model is robust yet interpretable.

With this report, we will be gauging the impact of key variables that include debt consolidation, employment length, and credit age in loan defaults. We will also show

how time and geographic variation may affect our model performance by adding these factors into our mixed-effects models.

Available Data

- The training data has 22457 rows and 18 columns.
- The testing data has 7474 rows and 18 columns.
- The validation data has 7494 rows and 18 columns.
- The extended version of the data has 37425 rows and 22 columns.

annual_inc: The self-reported annual income provided by the borrower during registration.

credit_age_yrs: The age of the oldest credit line in years (years since the first credit line was opened).

deliq_2yrs: The number of 30+ days past-due incidences of delinquency in the borrower's credit file for the past 2 years. Since this is historical data of the last two years, it could be used as a good indicator of the borrower's ability to repay the loan.

dti: A ratio calculated using the borrower's total monthly debt payments on the total debt obligations, excluding mortgage and the requested LC loan, divided by the borrower's self-reported monthly income. As this is a ratio of the borrower's monthly debt to income, and it does not include the mortgage, it could be a good indicator of the borrower's ability to repay the loan.

earliest_cr_line: The month the borrower's earliest reported credit line was opened. This could be a good indicator of the borrower's credit history.

emp_length: Employment length in years. Possible values are between less than 1 year and 10+ years. This could be a good indicator of the borrower's ability to repay the loan.

home_ownership: The homeownership status provided by the borrower during registration. Our data has three.

There are three categories: RENT, MORTGAGE, and OWN. This could be a good indicator of the borrower's ability to repay the loan.

inq_last_6mths: The number of inquiries in the past six months (excluding auto and mortgage inquiries). This could be a good indicator of the borrower's ability to repay the loan. Inquiries in this context are the number of times the borrower has applied for credit in the last six months.

int_rate: The interest rate on the loan. This could be a good indicator of the borrower's ability to repay the loan.

issue_d: The month in which the loan was funded. Only available in the extended version data.

loan_amnt: The listed amount of the loan applied for by the borrower. This could be a good indicator of the borrower's ability to repay the loan.

pub_rec: Number of derogatory public records.

Purpose: A category provided by the borrower for the loan request. This can be used in this analysis to see if the purpose of the loan affects the repayment rate.

revol_bal: Total credit revolving balance. A revolving balance is the amount of money owed to a credit card account at the end of a billing cycle. This could be a good indicator of the borrower's ability to repay the loan.

revol_util: Revolving line utilisation rate, or the amount of credit the borrower uses relative to all available revolving credit.

Term: The number of payments on the loan. Values are in months and can be either 36 or 60. This could be a good indicator of the borrower's ability to repay the loan.

total_acc: The total number of credit lines currently in the borrower's credit file. This could be a good indicator of the borrower's ability to repay the loan.

open_acc: The number of open credit lines in the borrower's credit file. This could be a good indicator of the borrower's ability to repay the loan.

To avoid multicollinearity, we will create a new variable called **close_acc**, which is the difference between **total_acc** and **open_acc**.

```
# add close accounts
extend_version_data$close_acc <- extend_version_data$total_acc -
extend_version_data$open_acc
...
# drop total_acc column
extend_version_data <- extend_version_data[, !colnames(extend_version_data)
%in% c("total_acc")]
...
```

verification_status: Indicates if income was verified by LC, not verified, or if the income source was verified. This could be a good indicator of the borrower's ability to repay the loan.

addr_state: The state provided by the borrower in the loan application. This could be a good indicator of the borrower's ability to repay the loan.

zip_code: The first three numbers of the zip code provided by the borrower in the loan application.

Data Preparation

- **Convert categorical variables to factors:**
Addressed in the section where categorical variables like `repay_fail`, `term`,

home_ownership, and verification_status are converted to factors for all datasets.

```
training_data$repay_fail <- factor(x = training_data$repay_fail)
...
# Applies to several other variables like term, purpose, home_ownership.
```

Removing invalid values:

Rows with invalid values (e.g., home_ownership = NONE and missing emp_length values) are removed.

```
extend_version_data <- extend_version_data[extend_version_data$home_ownership
!= "NONE", ]
...
# Same process for employment length, removing rows where emp_length = "n/a".
```

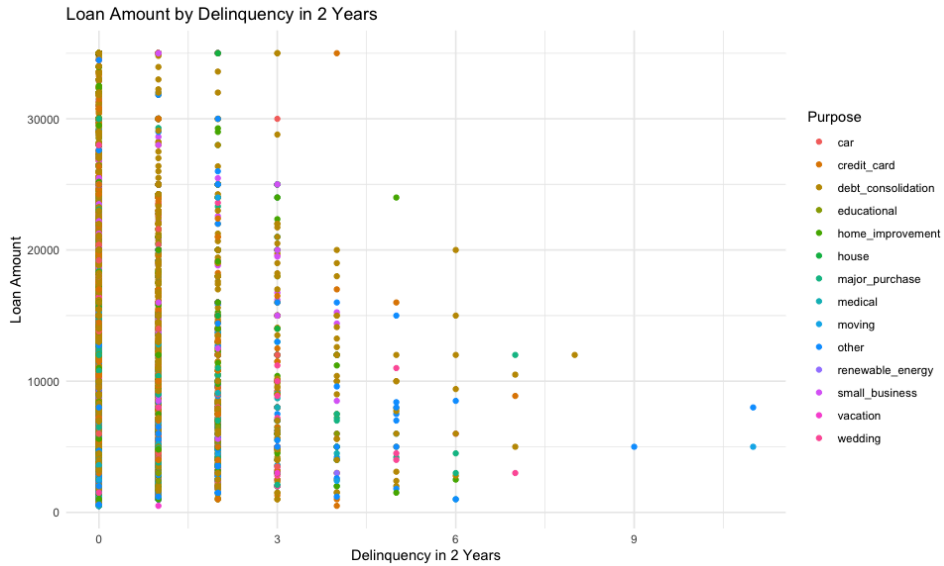
Date data:

The date data requires conversion to a Date object and correction of years in the 2068-2099 range.

```
# Convert to Date format (assume the 1st day of the month)
extend_version_data$earliest_cr_line <- as.Date(paste("01",
extend_version_data$earliest_cr_line), format = "%d %b-%y")
# Check and correct years that are incorrectly in 2068-2099 range
extend_version_data$earliest_cr_line <-
ifelse(year(extend_version_data$earliest_cr_line) > 2024,
extend_version_data$earliest_cr_line - years(100),
extend_version_data$earliest_cr_line)
# Convert to Date object again (ifelse returns characters, so reconvert)
extend_version_data$earliest_cr_line <-
as.Date(extend_version_data$earliest_cr_line)
extend_version_data$issue_d <- as.Date(paste("01",
extend_version_data$issue_d), format = "%d %b-%y")
```

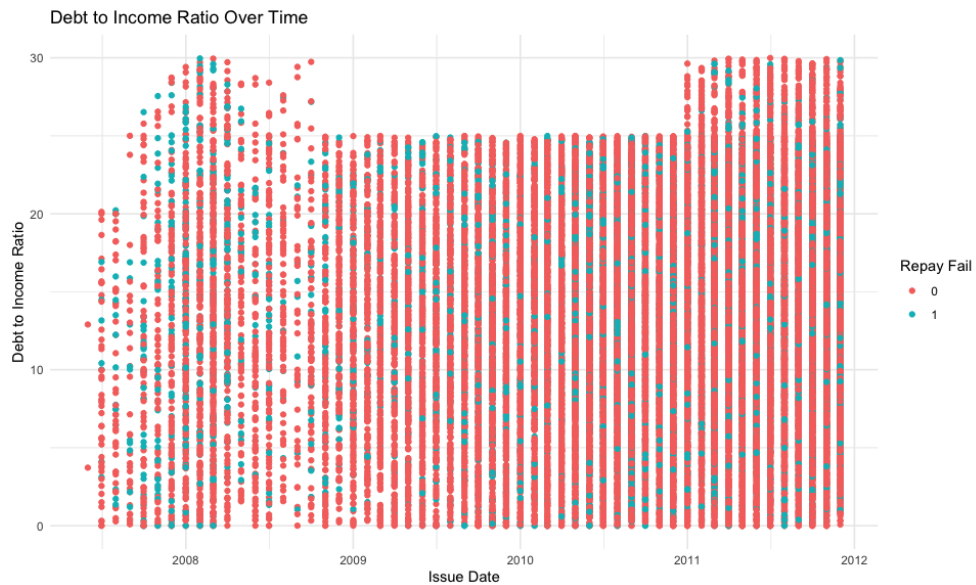
3. Visualisation of Variable Relationships

- **Loan Amount by Delinquency in 2 Years and Purpose:** Visualizes how delinq_2yrs (delinquency in 2 years) relates to loan amounts and purposes.



This plot above allows for the observation of how delinquency correlates with the loan amount across different loan purposes.

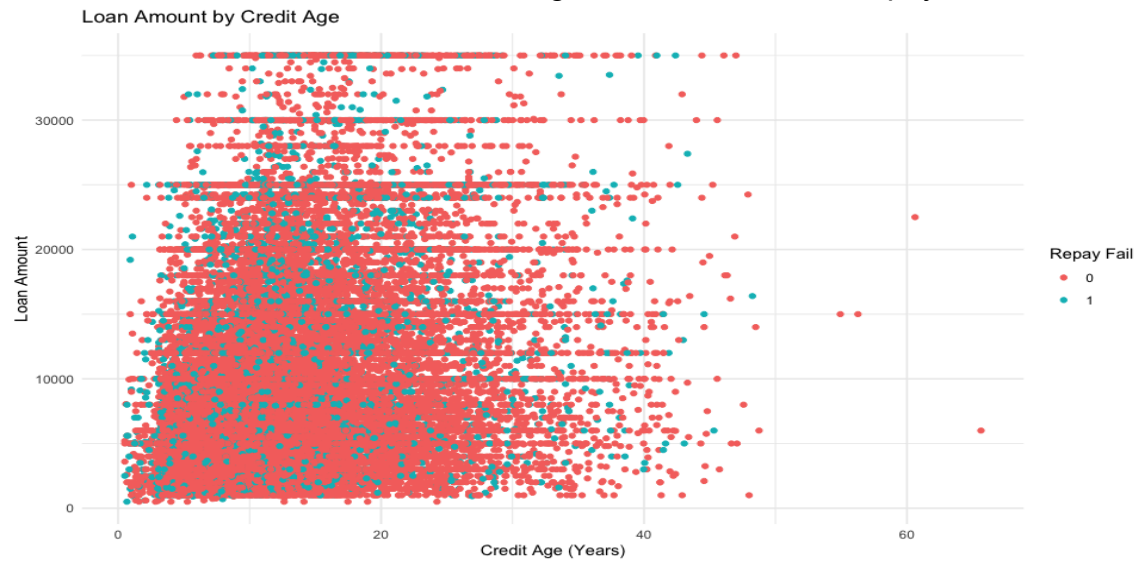
- Debt-to-Income Ratio over Time and Repay Fail:** This shows how dti (debt-to-income ratio) changes over time and how it correlates with loan repayment failures.



Debt to Income Ratio Over Time

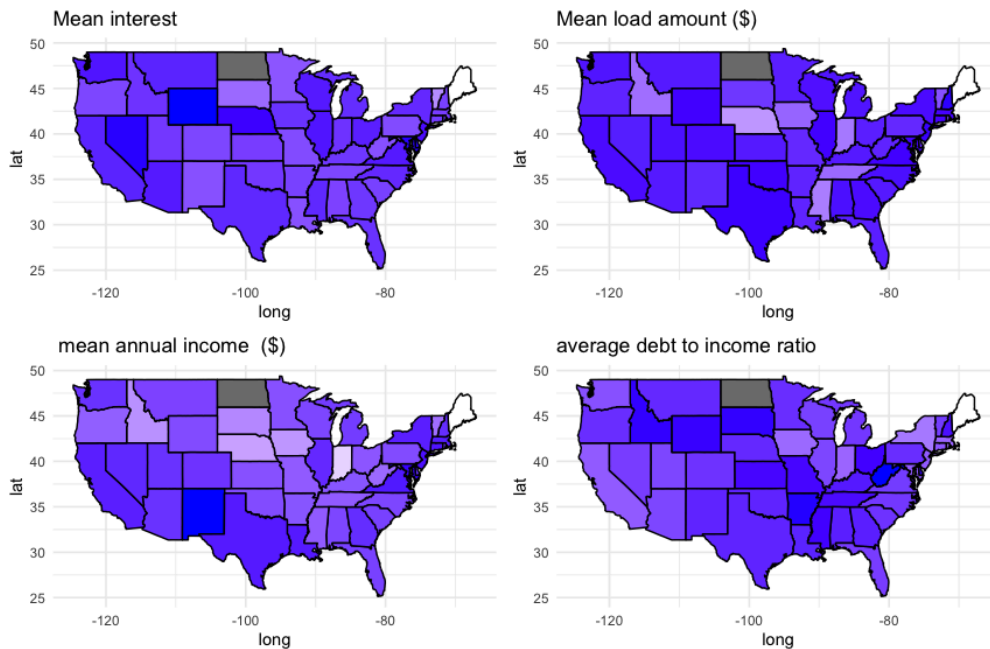
This chart highlights potential trends in dti and repayment success/failure, showing how borrowers' debt loads have changed over time.

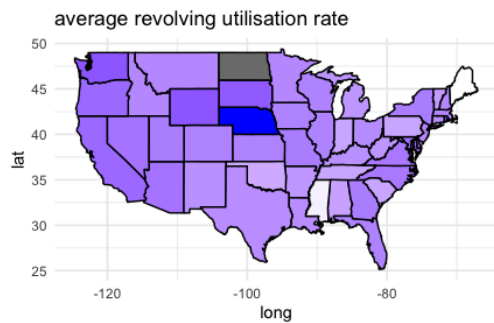
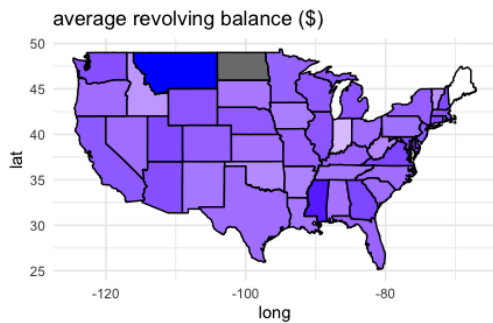
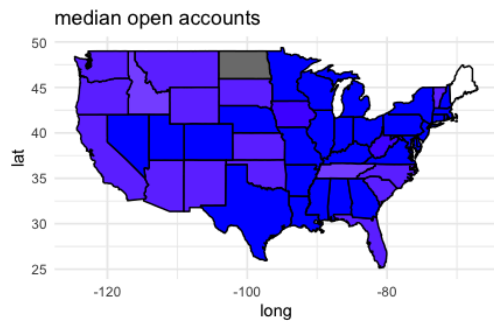
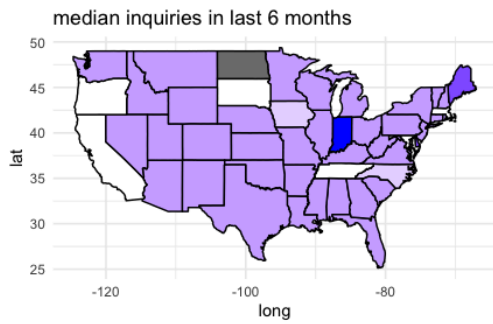
- **Credit Age and Loan Amount by Repay Fail:** Explores the relationship between the borrower's credit age, loan amount, and repayment status.

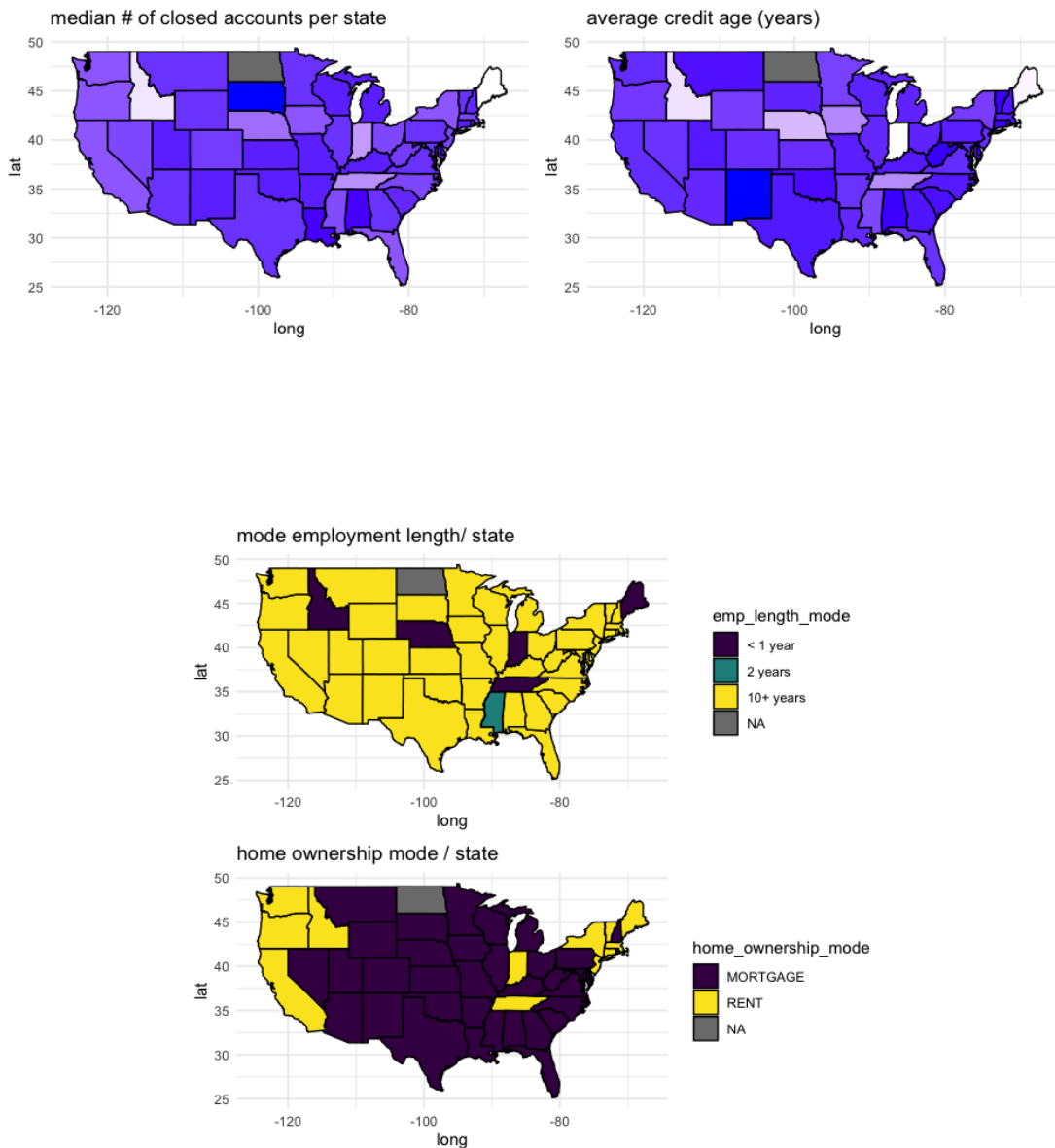


The plot above shows a slight positive correlation between credit age and loan amounts, with distinctions made between those who repaid and those who failed to repay.

States Data Visualisation



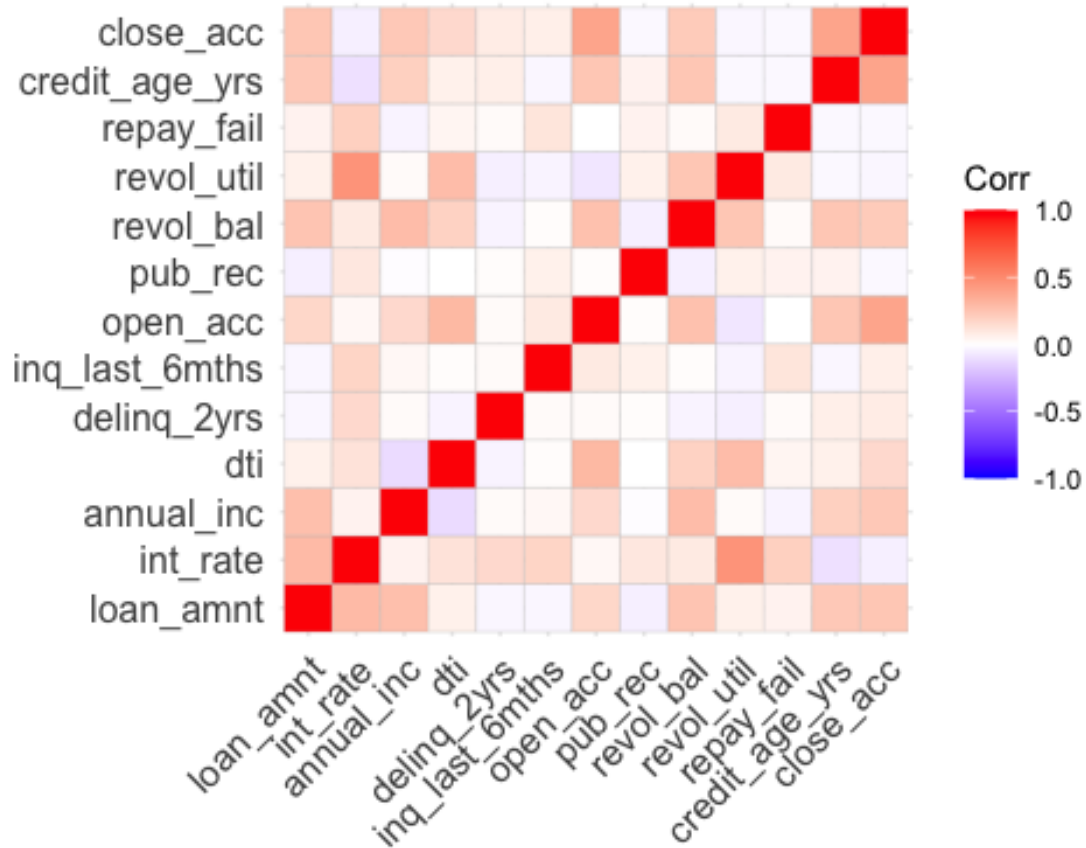




There appear to be multiple relationships between the geographic location and the loan data. For example, the mean interest rate per state is highest in Nevada, and the highest mean annual income is in New Mexico. The model employment length is 10+ years in most states, except for states like Idaho and Nebraska, which have a mode of less than 1 year in this dataset.

Correlation Analysis

- **Correlation Matrix for Numeric Variables:** A correlation matrix is generated to identify correlations between variables like interest rate, revolving utilisation, credit age, etc.



The correlation matrix reveals strong relationships between several numeric variables, such as a positive correlation between open accounts and total accounts, revolving utilisation, and interest rate.

Section 1: Generalised Linear Model for Loan Repayment Failure Prediction

Question 1

How does your new model perform compared to the one you used previously? How can it be expected to perform on new loan applications? For this, you must use the training and validation data from the previous benchmark.

Question 2

What are the important variables in this model, and how do they compare to traditionally important variables for predicting credit risk in the banking sector? One regulatory requirement for lenders is clearly explaining how a loan application is assessed. To demonstrate to management that this can be achieved based on your model, clearly interpret all covariates in your model regarding their effect on predicting credit risk, including uncertainty.

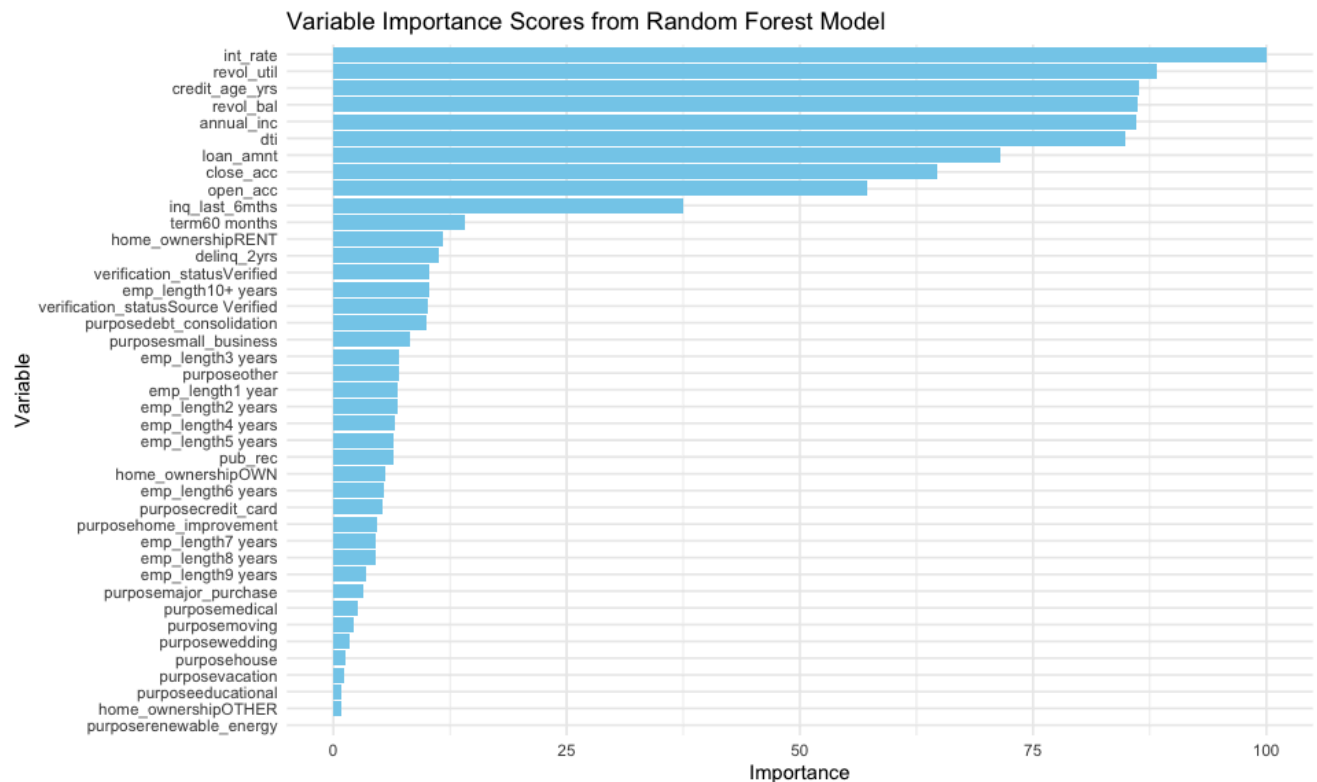
Data:

Given the large number of variables and data points, we will use a random forest model to select important variables and predict the target variable. This is done by training a generalised linear model with all variables up to 2-way interactions proved computationally expensive, hence a random forest model to select influential variables and then train a generalised linear model with the selected variables with stepwise selection.

Random Forest Model

```
set.seed(9940669) # For reproducibility
# Define a grid of hyperparameters to tune
tune_grid <- expand.grid(mtry = c(2, 4, 6, 8))
# Set up cross-validation method
control <- trainControl(method = "cv", number = 5)
# Train the random forest model using the grid of hyperparameters
rf_grid_search <- train(repay_fail ~ loan_amnt + term + int_rate + emp_length
+ home_ownership + annual_inc + verification_status + purpose + dti +
delinq_2yrs + inq_last_6mths + open_acc + pub_rec + revol_bal + revol_util +
credit_age_yrs + close_acc,
                        data = training_data,
                        method = "rf",
                        trControl = control, # Cross-validation method
                        tuneGrid = tune_grid,
                        ntree = 500)
# some of these code snippets were generated and debugged using Chatgpt
```

The importance of variables is shown in the following plot:



Based on the random forest model, the most important variables for predicting loan repayment failure are the interest rate, revolving utilisation rate, credit age, revolving balance, annual income, dti, loan amount, number of close accounts and number of open accounts with an importance score of 50 or higher.

The number of inquiries in the last 6 months, loan term, home ownership status, verification status and the number of delinquencies in the last 2 years are somewhat important variables for predicting loan repayment failure, with importance scores between 10 and 40.

The majority of values of employment length are not important for predicting loan repayment failure, with importance scores below 10, except for the 10+ years category, which has an importance score of 10. Similarly, All loan purposes are not important for predicting loan repayment failure, with importance scores below 10, except for the small purpose of debt consolidation, which has an importance score of 10.

Based on these results, the saturated model will be reduced to include only the important variables with an importance score of 10 or higher and two-way interactions of variables with a score of 50 or higher. Note that these numbers are arbitrary and only used as a starting point for the model reduction.

Therefore, employment length of 10+ years and loan purpose of debt consolidation will be encoded as their variables in the model.

```
# encode employment length over 10 years as its own variable
training_data$emp_length_10 <- ifelse(training_data$emp_length == "10+
years", 1, 0)

# purpose of dept consolodation as its own variable
training_data$purpose_debt_consolidation <- ifelse(training_data$purpose ==
"debt_consolidation", 1, 0)
```

This will reduce the 'saturated' model from:

```
full_logistic_model <- glm(repay_fail ~ (loan_amnt + term + int_rate +
  emp_length + home_ownership + annual_inc + verification_status +
  purpose + dti + delinq_2yrs + inq_last_6mths + open_acc + pub_rec +
  revol_bal + revol_util + credit_age_yrs + close_acc)^2,
  data = training_data, family = "binomial")
```

to:

```
reduced_logistic_model <- glm(repay_fail ~ home_ownership + inq_last_6mths +
  term + delinq_2yrs + verification_status +
  emp_length_10 + purpose_debt_consolidation +
  (int_rate + revol_util + credit_age_yrs +
  annual_inc +
  dti + loan_amnt + close_acc + open_acc )^2,
  data = training_data, family = "binomial")
```

The AIC of the full model is 18075 and the AIC of the reduced model is 17717. The reduced model has a lower AIC value, indicating it fits the data better than the full model.

Stepwise Selection

The reduced model will be used as the starting point for stepwise selection to identify the best model based on the AIC value. This will reduce the number of variables in the model, improve its interpretability, and reduce the risk of overfitting and complexity.

```
step_model <- stepAIC(null_model,
  scope = formula(reduced_logistic_model),
  direction = "both",
  trace = 0)
```

Coefficients:

##	Estimate	Std. Error	z value	Pr(> z)	
## (Intercept)	-3.822e+00	1.731e-01	-22.077	< 2e-16	***
## int_rate	1.588e-01	1.251e-02	12.695	< 2e-16	***

```

## inq_last_6mths          1.445e-01  1.153e-02  12.534 < 2e-16 ***
## annual_inc             -1.078e-05  1.632e-06  -6.607 3.93e-11 ***
## term60 months          4.028e-01  4.791e-02   8.407 < 2e-16 ***
## revol_util              1.384e+00  2.899e-01   4.773 1.82e-06 ***
## emp_length_101          1.290e-01  4.660e-02   2.769 0.00563 **
## purpose_debt_consolidation1 -7.093e-02  4.002e-02  -1.773 0.07629 .
## loan_amnt               1.481e-05  7.529e-06   1.967 0.04916 *
## int_rate:revol_util     -7.934e-02  2.034e-02  -3.901 9.58e-05 ***
## annual_inc:revol_util    6.007e-06  2.063e-06   2.912 0.00359 **
## revol_util:loan_amnt    -2.799e-05  1.045e-05  -2.678 0.00740 **
## annual_inc:loan_amnt     8.862e-11  5.457e-11   1.624 0.10441
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 19004  on 22456  degrees of freedom
## Residual deviance: 17690  on 22444  degrees of freedom
## AIC: 17716
##
## Number of Fisher Scoring iterations: 5

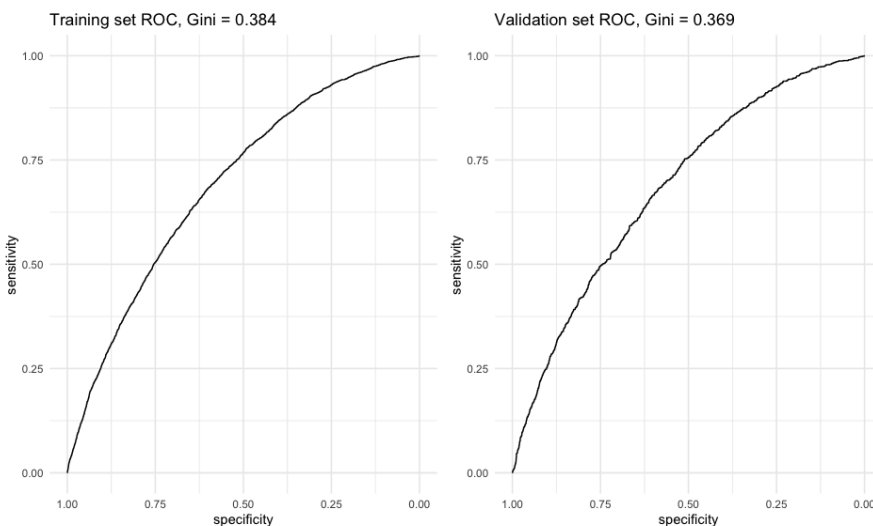
```

The stepwise selection model has the lowest AIC value, indicating that it is the best model among the three in terms of goodness of fit and complexity.

The dispersion parameter of the forward model is 1, which indicates that the model is well-calibrated.

Model Evaluation

Based on feedback from stakeholders, the model will be evaluated using the validation data set. The model will be used to predict the probability of loan repayment failure for each loan application in the validation data set. The model will then be evaluated using the area under the ROC curve (AUC) and the Gini coefficient. The Gini coefficient



measures the model's ability to discriminate between good and bad loans, with a higher value indicating better performance.

ROC Curves for the Training and Validation Data Sets with Gini Scores

Comparison of the models

##	Model	Training_gini	Validation_gini	aic	num_variables
## 1	Reduced Logistic	39.1	37.2	17717	47
## 2	Stepwise Selection	38.4	36.9	17716	13
## 3	Full Logistic	49.4	34.4	18075	735
## 4	Benchmark	11.4	11.0	NA	NA

Compared to the Benchmark model, all the models developed outperformed it in terms of the Gini score. The full logistic model has the highest Gini score on the training data. on the validation data, the reduced logistic model has the highest Gini score, followed by the stepwise selection model. The full logistic model has the lowest Gini score of the 3 GLMs on the validation data, indicating that it is overfitting the training data.

Given how close the AIC scores and Gini scores of the reduced logistic model and the stepwise selection model are on the validation data and that the stepwise selection model has fewer variables, it is the best model among the three in terms of goodness of fit and complexity.

Interpretation of the Model

To improve interpretability, the coefficients of the stepwise selection model will be exponential to show the effect of each variable. The load amount and annual income will be scaled to 10 thousand, and the revolving utilisation rate will be scaled by 100 to make the coefficients more interpretable.

```
training_data$loan_amnt_10k <- training_data$loan_amnt/10000
training_data$annual_inc_10k <- training_data$annual_inc/10000
training_data$revol_util_scaled <- training_data$revol_util*100
##                               exp_Coefficient p_value      Effect
## int_rate                     1.17215755 <0.0001 + 17.21575 %
## inq_last_6mths               1.15551623 <0.0001 + 15.55162 %
## annual_inc_10k               0.89779445 <0.0001 - 10.22056 %
## term36 months               0.02187668 <0.0001 - 97.81233 %
## term60 months               0.03272793 <0.0001 - 96.72721 %
## revol_util_scaled            1.01393450 <0.0001 + 1.39345 %
## emp_length_101              1.13770344 0.00563 + 13.77034 %
## purpose_debt_consolidation1 0.93152329 0.07629 - 6.84767 %
## loan_amnt_10k               1.15963612 0.04916 + 15.96361 %
## int_rate:revol_util_scaled   0.99920692 1e-04 - 0.07931 %
## annual_inc_10k:revol_util_scaled 1.00060090 0.00359 + 0.06009 %
## revol_util_scaled:loan_amnt_10k 0.99720499 0.0074 - 0.2795 %
## annual_inc_10k:loan_amnt_10k 1.00890100 0.10441 - 0.8901 %
##                               conf_lower    conf_upper
## int_rate                     + 14.38482 % + 20.13618 %
```

## inq_last_6mths	+ 12.97475 %	+ 18.19884 %
## annual_inc_10k	- 13.06633 %	- 7.32479 %
## term36 months	- 98.44585 %	- 96.93606 %
## term60 months	- 97.72487 %	- 95.31307 %
## revol_util_scaled	+ 0.82102 %	+ 1.97356 %
## emp_length_101	+ 3.79842 %	+ 24.60307 %
## purpose_debt_consolidation1	- 13.8778 %	+ 0.74952 %
## loan_amnt_10k	+ 0.04421 %	+ 34.39396 %
## int_rate:revol_util_scaled	- 0.11927 %	- 0.0396 %
## annual_inc_10k:revol_util_scaled	+ 0.01972 %	+ 0.10061 %
## revol_util_scaled:loan_amnt_10k	- 0.48302 %	- 0.07448 %
## annual_inc_10k:loan_amnt_10k	- 0.20399 %	+ 1.9534 %

The significant variables in the selected model are Interest rate, inquiries in the last 6 months, Annual income, loan term (36 or 60 months), Revolving utilisation rates and loan amount.

The significant interactions are:

- Interest rate and revolving utilisation rate.
- Annual income and revolving utilisation rate.
- Revolving utilisation rate and loan amount.
- Annual income and loan amount.

In the literature, Dm & Mm (2018) developed a logistic regression model to predict Individual Loan Default rates, where only the number of credits taken, existing credits, and delayed payments (similar to delinquencies) had a significant p-value (alpha of 0.05).

Li & Wu (2024) developed a logistic regression model to predict loan default rates. The variables with a significant p-value were age, working years and current job years (similar to working length in our data), income, marriage status, car ownership, and renting status (similar to homeownership in our data).

Only the annual income and employment length are the common significant variables in the designed model compared to the 2 models cited.

Effect of each variable:

Interest rate has a p-value of less than 0.0001, indicating that it is a significant predictor of loan repayment failure. For every 1% increase in the interest rate, the odds of loan repayment failure increase by 17%, with a 95% confidence interval of 20% to 14% risk increase.

Inquiries in the last 6 months have a p-value of less than 0.0001, indicating that it is a significant predictor of loan repayment failure. For every additional inquiry in the last 6 months, the odds of loan repayment failure increase by approximately 16%. The number of inquiries in the last 6 months has a 95% confidence interval of 18% to 13% risk increase.

Annual income has a p-value of less than 0.0001, indicating that it is a significant predictor of loan repayment failure. For every \$10,000 increase in annual income, the odds of loan repayment failure decrease by 10%. The annual income has a 95% confidence interval of 13% to 7% risk decrease.

The term of the loan has a p-value of less than 0.0001, indicating that it is a significant predictor of loan repayment failure. Loans with a term of 60 months decrease the odds of loan repayment failure by 97%, and a 95% confidence interval of 98% to 95% chance decrease in loan repayment failure. A term of 36 months decreases the odds of loan repayment failure by 98%, and a confidence interval of 98% to 97% decreases loan repayment failure.

The revolving utilisation rate has a p-value of less than 0.0001, indicating that it is a significant predictor of loan repayment failure. For every 1% increase in the revolving utilisation rate, the odds of loan repayment failure increase by 1%. The revolving utilisation rate has a 95% confidence interval of 2% to 1% risk increase.

Employment length of 10+ years has a p-value of 0.00563, indicating that it is a significant predictor of loan repayment failure. Loan applicants with employment length of 10+ years have a 14% higher odds of loan repayment failure than those with employment length of less than 10 years. The employment length of 10+ years has a 95% confidence interval of 24.6% to 3.8% risk increase, showing that the effect of the employment length of 10+ years on loan repayment failure has uncertainty.

The loan amount has a p-value of 0.049, indicating that it is a significant predictor of loan repayment failure. For every \$10,000 increase in the loan amount, the odds of loan repayment failure increase by 16%. The loan amount has a 95% confidence interval of 0.04% to 34% risk increase, showing that the effect of the loan amount on loan repayment failure has high uncertainty.

Interest rate and revolving utilisation rate have a p-value of less than 0.0001, indicating that it is a significant predictor of loan repayment failure. The effect of the interaction is multiplicative, meaning that the effect of the interest rate on loan repayment failure is amplified by the revolving utilisation rate. For every 1% increase in the interest rate and revolving utilisation rate, the odds of loan repayment failure decrease by Approximately 0.1%, with a 95% confidence interval of 0.12% to 0.04% risk decrease.

Annual income and revolving utilisation rate has a p-value of less than 0.004, indicating that it is a significant predictor of loan repayment failure. For every \$10,000 increase in annual income and 1% increase in the revolving utilisation rate, the odds of loan repayment failure increase by 0.06%, with a 95% confidence interval of 0.1% to 0.02% risk increase.

Revolving utilisation rate and loan amount has a p-value of less than 0.0074, and for every 1% increase in the revolving utilisation rate and \$10,000 increase in the loan

amount, the odds of loan repayment failure decreases by 0.28%, with a 95% confidence interval of 0.48% to 0.07% risk decrease.

Insignificant variables

The purpose of debt consolidation has a p-value of 0.07629, indicating that it is not a significant predictor of loan repayment failure. Loan applicants with the purpose of debt consolidation have a 6.8% lower odds of loan repayment failure than those with other purposes. The purpose of debt consolidation has a 95% confidence interval of 13.9% decrease to 0.75% increase in loan repayment failure, showing that the effect of the purpose of debt consolidation on loan repayment failure has a higher level of uncertainty.

Annual income and loan amount have a p-value of 0.10441, indicating that it is not a significant predictor of loan repayment failure. For every \$10,000 increase in annual income and loan amount, the odds of loan repayment failure decrease by approximately 1%, with a 95% confidence interval of 1.95% increase to 0.2% decrease in loan repayment failure, showing that the effect of the annual income and loan amount on loan repayment failure has a higher level of uncertainty.

Section 2: Mixed Effects Models

After management consultations with experts, it has been decided to use the extended version of the data to address the following questions:

3. Can accounting for this variation (e.g., state/zip-code and time) improve performance benchmarks?
4. Are there any surprising differences in variables important for predicting credit risk? This is again essential for regulations.
5. Does credit risk change over time or between states? The bank has not previously investigated this, and the results may inform modified loan policies in the future.

Mixed Effects Model

The initial model used to analyse the effect of Zip Code and Issue Dates was:

```
state_time_model <- lmer(data = extended_data,
  repay_fail ~ home_ownership +
    inq_last_6mths + term + delinq_2yrs +
    verification_status + emp_length_10 +
    purpose_debt_consolidation + (int_rate +
      revol_util_scaled
```

```

+
credit_age_yrs +
annual_inc_10k +
dti +
loan_amnt_10k +
close_acc +
open_acc)^2 +
(1 | zip_code) + (1 | issue_d))
AIC(state_time_model)
## [1] 27431.48

```

The AIC of the mixed effects model is higher than that of the initial model. This is to be expected, as the increased number of interactions between each variable increases the complexity of the model. Thus, the AIC value is raised.

When analysing the significance of each covariate in the model, it is observed that a select number of covariates, such as `revol_util`, `annual_inc` and `loan_amnt`, all become insignificant, see Appendix A.

To understand whether removing the covariates from the final model improves benchmarks, the model below was built to test this.

```

state_time_model_2 <- lmer(data = extended_data,
  formula <- repay_fail ~ home_ownership +
    inq_last_6mths + term + delinq_2yrs +
    verification_status + emp_length_10 +
    purpose_debt_consolidation + (int_rate +
      credit_age_yrs +
      dti +
      close_acc +
      open_acc)^2 +
    (1 | zip_code) + (1 | issue_d))

```

Checking the AIC of the new mixed effects model.

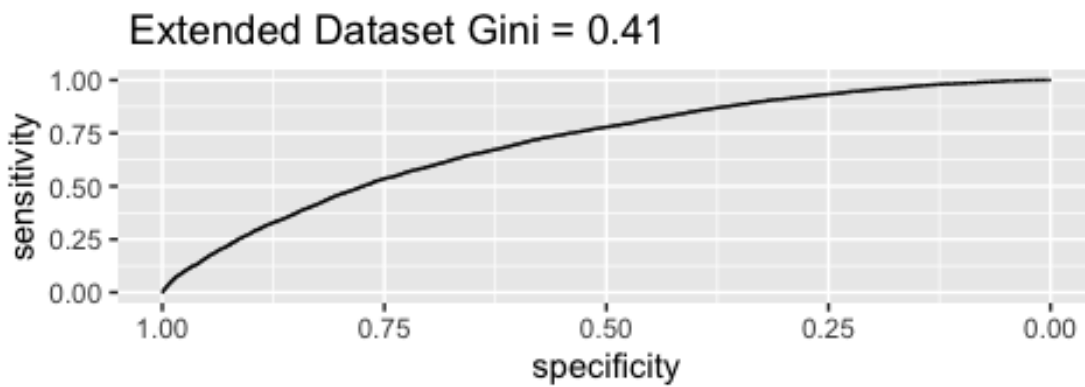
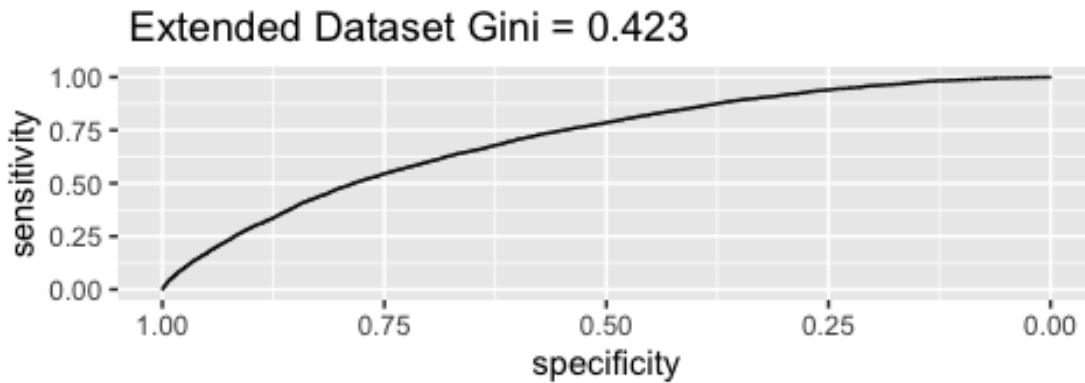
```

AIC(state_time_model_2)
## [1] 27218.43

```

The AIC of the second model is lower than the original model, indicating the less complex model is potentially a better fit than the initial mixed effects model.

Performance benchmarks, such as Gini Scores and the AUC/ROC should be analysed to check for the model fit to justify whether the more complex model can fit the data better.



When comparing the Gini Scores, overall, it is noted that accounting for the variation between jurisdictions and over time in the credit risk model does improve performance benchmarks to a minor extent.

However, while the second model with fewer covariates, see Appendix B for model summary, has a better AIC score, the Gini Score indicates greater heterogeneity in the initial mixed effects model. Having greater heterogeneity in the model is important, as it better distinguishes customers who are likely able to pay off their loans or default on them.

Comparing Mixed Effects Model and Generalised Linear Model

```
##           Model Gini
## 1           GLM 37.8
## 2 Mixed Effects 41.4
```

To evaluate model performance, we compared the Gini scores of the mixed effects model and the generalised linear model (GLM) using identical parameters. The mixed effects model achieved a Gini score 41.4, outperforming the GLM score 37.8. This suggests that the mixed effects model more effectively differentiates between high-risk and low-risk loans.

Effects of Time and Location on Credit Risk

We next explored the impact of the loan issue date and geographic location (zip code) on credit risk, measured by Gini scores and AIC values. Results are summarised below:

```
##           Model Gini   AIC
## 1           GLM 37.8 29467
## 2 Mixed Effects Reduced 41.4 26935
## 3 Mixed Effects Zip code 40.1 26982
## 4 Mixed Effects Date 38.5 27000
```

Including the zip code as a random effect yielded a higher Gini score than using the issue date, suggesting that geographic location has a more significant impact on credit risk than the issue date alone.

Accounting for State and Zip Code in the Model

To further assess the geographic impact, we introduced a combined variable for state and zip code by introducing a variable **sample** to account for the state and zip code in the model

```
extended_data$sample <-
  factor(as.factor(extended_data$addr_state):extended_data$zip_code)

state_zip_model_select_reduced_sample <- lmer(data = extended_data,
  repay_fail ~ int_rate + inq_last_6mths + annual_inc_10k + term +
  revol_util_scaled + emp_length_10 + purpose_debt_consolidation +
  int_rate:revol_util_scaled + (1 | sample) )
```

Scores:

```
##           Model   Gini   AIC
## 1           GLM 37.800 29467
## 2 Mixed Effects Reduced 41.400 26935
## 3 Mixed Effects Zip code 40.100 26982
## 4 Mixed Effects Date 38.500 27000
```

```
## 5           Mixed Effects Sample(state:zip) 40.115 26981
## 6 Mixed Effects Sample(state:zip, issue date) 41.454 26934
```

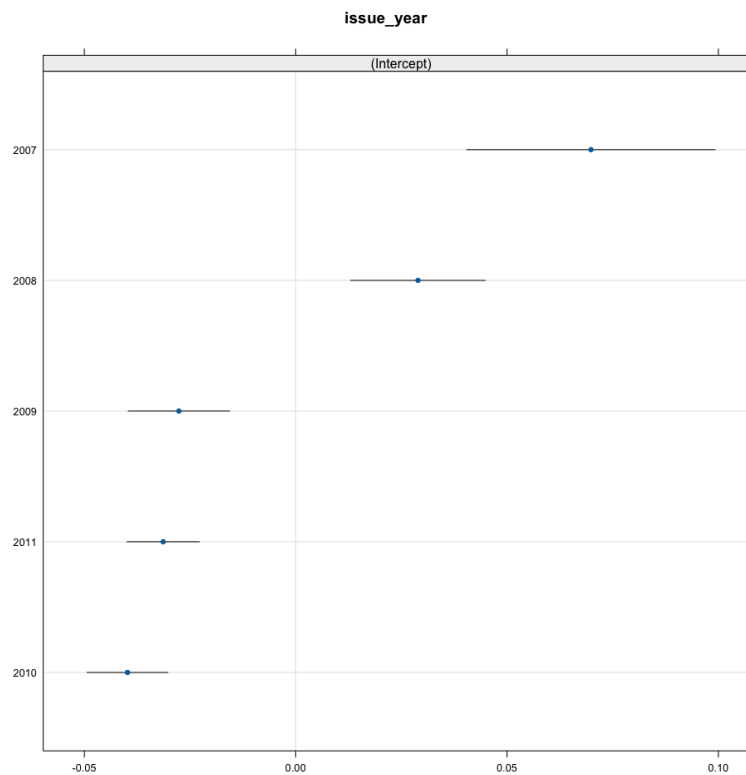
The mixed effects model with the sample variable (state:zip code). Issue date has the highest Gini score, indicating that it is the best model among the four regarding goodness of fit and complexity. The mixed effects model with the sample variable also has the lowest AIC value, indicating that it is the best model among the four regarding goodness of fit and complexity.

Visualisation and Interpretation of Final Model

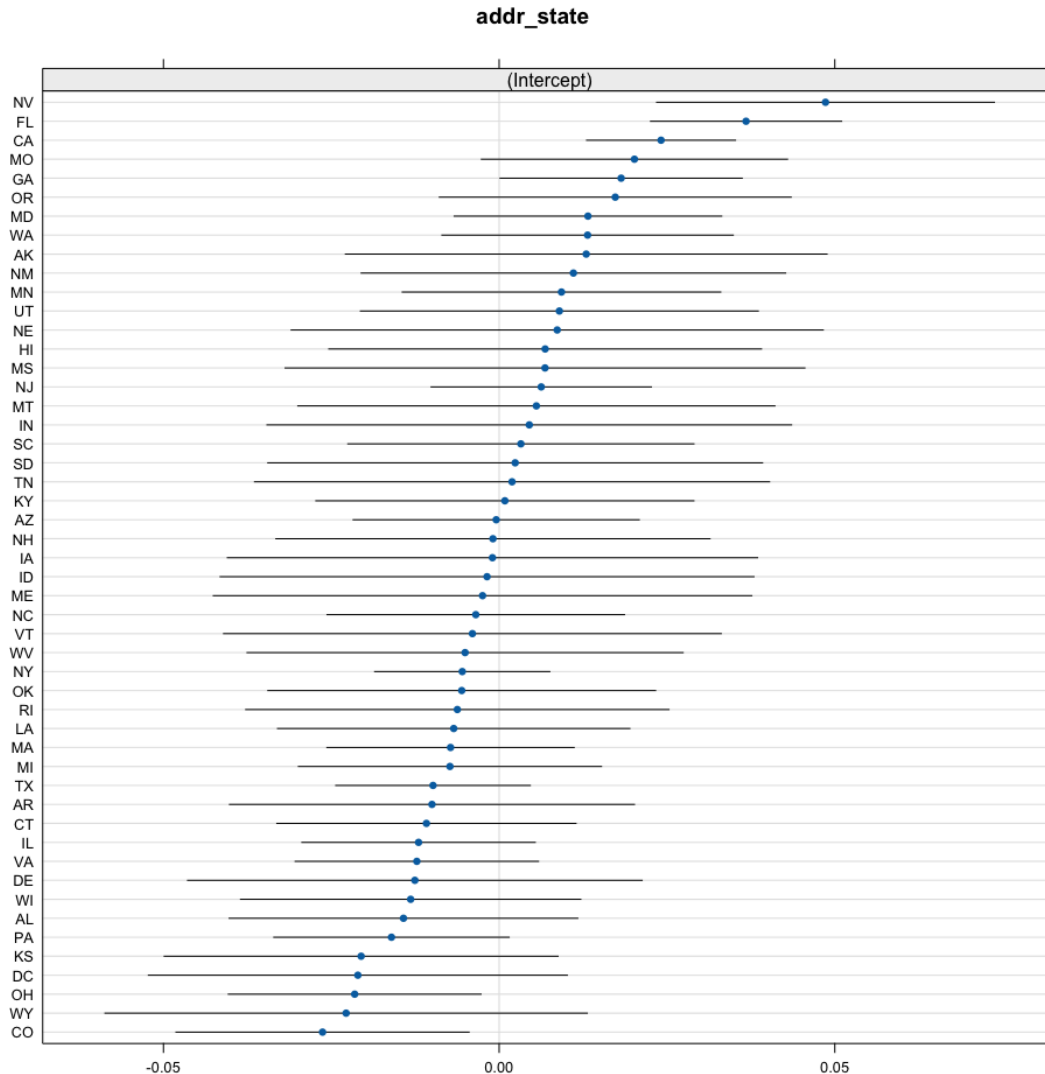
To visualize the effect of loan issue year and state on credit risk, we built a model with these two variables as random effects:

```
year_levels <- as.character(2007:2011)
extended_data$issue_year <- factor(x = year(extended_data$issue_d), levels =
year_levels)
fit <- lmer(data = extended_data,
            repay_fail ~ int_rate + inq_last_6mths +
annual_inc_10k +
            term + revol_util_scaled + emp_length_10 + purpose_debt_consolidation +
            loan_amnt_10k + int_rate:revol_util_scaled +
annual_inc_10k:revol_util_scaled +
            revol_util_scaled:loan_amnt_10k + annual_inc_10k:loan_amnt_10k + (1 |
issue_year) + (1 | addr_state) )
```

Interpretation of the random effects



Random effect on year: The years of 2007 and 2008 have a positive intercept effect on credit risk, meaning loans in these two years are more likely to fail than loans in other years, where the years of 2009, 2010, and 2011 have a negative effect. Furthermore, none of the intercept points and their deviance lines are close to the zero line, indicating that the effect of the year on credit risk is significant and has a large effect on credit risk.



The random effect of the state: The states of NV (Nevada), FL (Florida), CA (California) and GA (Georgia) all do not overlap zero and have a positive intercept, indicating that loans in these states are more likely to fail than loans in other states. CO (Colorado) and OH (Ohio) are the only states with a negative intercept that do not overlap zero, indicating that loans in these states are less likely to fail than loans in other states.

Interpretation of the fixed effect

##	exp_Coefficient	Effect	conf_lower
## int_rate	1.0140071	+ 1.40071 %	+ 1.61902 %
## inq_last_6mths	1.0195160	+ 1.9516 %	+ 2.20419 %
## annual_inc_10k	0.9984992	- 0.15008 %	- 0.02114 %
## term36 months	0.9652351	- 3.47649 %	- 1.73877 %
## term60 months	1.0362235	- 3.62235 %	- 9.38173 %
## revol_util_scaled	1.0004705	+ 0.04705 %	+ 0.0934 %
## emp_length_101	1.0164221	+ 1.64221 %	+ 2.51161 %
## purpose_debt_consolidation1	0.9934169	- 0.65831 %	- 0.06817 %

## loan_amnt_10k	0.9943829 + 0.56171 % + 0.61263 %
## int_rate:revol_util_scaled	0.9999985 - 0.00015 % - 0.00344 %
## annual_inc_10k:revol_util_scaled	0.9999763 + 0.00237 % + 3e-04 %
## revol_util_scaled:loan_amnt_10k	1.0000564 - 0.00564 % - 0.02398 %
## annual_inc_10k:loan_amnt_10k	0.9998795 - 0.01205 % - 0.04754 %
##	conf_upper p_value
## int_rate	+ 1.18276 % <0.0001
## inq_last_6mths	+ 1.70472 % <0.0001
## annual_inc_10k	- 0.27893 % 0.02261
## term36 months	- 8.32783 % 0.20801
## term60 months	- 1.72439 % 0.21539
## revol_util_scaled	+ 7e-04 % 0.04664
## emp_length_101	+ 0.77739 % 0.00018
## purpose_debt_consolidation1	+ 1.38128 % 0.07602
## loan_amnt_10k	+ 1.72449 % 0.34754
## int_rate:revol_util_scaled	- 0.00373 % 0.93549
## annual_inc_10k:revol_util_scaled	+ 0.00444 % 0.0251
## revol_util_scaled:loan_amnt_10k	- 0.01268 % 0.54623
## annual_inc_10k:loan_amnt_10k	+ 0.07153 % 0.69163

Significant variables:

- The interest rate has a p-value of less than 0.0001. For every 1% increase in the interest rate, the odds of loan repayment failure increase by 1.4%, with a 95% confidence interval of 1.6% to 1.1% risk increase.
- Inquiries in the last 6 months have a p-value of less than 0.0001. For every additional inquiry in the last 6 months, the odds of loan repayment failure increase by 1.95%, with a 95% confidence interval of 2.2% to 1.7% risk increase.
- Annual income has a p-value of 0.02261. For every \$10,000 increase in annual income, the odds of loan repayment failure decreased by 0.15%, with a 95% confidence interval of 0.28% to 0.02% risk decrease.
- The revolving utilisation rate has a p-value of less than 0.04664. For every 1% increase in the revolving utilisation rate, the odds of loan repayment failure increase by 0.04705%, with a 95% confidence interval of 0.0934% to 7e-04% risk increase.
- emp_length_10 has a p-value of 0.00018. Loan applicants with employment lengths of 10+ years have 1.64% higher odds of loan repayment failure than those with less than 10 years. The employment length of 10+ years has a 95% confidence interval of 0.77739% to 0.77739% risk increase.
- The interaction term between annual income and revolving utilisation rate is Significant, with a p-value of 0.0251. Every \$10,000 increase in the loan amount and 1% increase in the utilisation rate is associated with a 0.00237% increase in repayment failure rate, with a 95% confidence interval of 3e-04% to 0.00444%

Increase. The loan term, the purpose of debt consolidation, the loan amount, and other interactions are insignificant in this model.

References

Dm, O., & Mm, M. (2018). Comparison of Accuracy of Support Vector Machine Model and Logistic Regression Model in Predicting Individual Loan Defaults. American Journal of Applied Mathematics and Statistics. <http://pubs.sciepub.com/ajams/6/6/8>

Li, H., & Wu, W. (2024). Loan default predictability with explainable machine learning. Finance Research Letters, 60, 104867. <https://doi.org/10.1016/j.frl.2023.104867>

Appendix A

```
summary(state_time_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: repay_fail ~ home_ownership + inq_last_6mths + term + delinq_2yrs
+
##      verification_status + emp_length_10 + purpose_debt_consolidation +
##      (int_rate + revol_util_scaled + credit_age_yrs + annual_inc_10k +
##      dti + loan_amnt_10k + close_acc + open_acc)^2 + (1 |
##      zip_code) + (1 | issue_d)
##      Data: extended_data
##
## REML criterion at convergence: 27331.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.4346 -0.5248 -0.3323 -0.1127  5.4235
##
## Random effects:
##      Groups   Name                Variance Std.Dev.
##      zip_code (Intercept) 0.0007765 0.02787
##      issue_d  (Intercept) 0.0009674 0.03110
##      Residual                0.1185861 0.34436
## Number of obs: 37425, groups:  zip_code, 830; issue_d, 55
##
## Fixed effects:
##
##              Estimate Std. Error      df t
value
## (Intercept)      -4.038e-02  2.616e-02  1.350e+04
-1.544
## home_ownershipOTHER      7.130e-02  3.155e-02  3.668e+04
2.260
## home_ownershipOWN       7.332e-03  7.268e-03  3.727e+04
```

1.009			
## home_ownershipRENT	9.702e-03	4.300e-03	2.419e+04
2.256			
## inq_last_6mths	1.989e-02	1.254e-03	3.113e+04
15.854			
## term60 months	6.366e-02	5.316e-03	1.718e+04
11.973			
## delinq_2yrs	3.600e-04	3.734e-03	3.723e+04
0.096			
## verification_statusSource Verified	-1.401e-03	4.865e-03	1.742e+04
-0.288			
## verification_statusVerified	-5.047e-04	4.888e-03	2.148e+04
-0.103			
## emp_length_101	1.737e-02	4.567e-03	3.731e+04
3.804			
## purpose_debt_consolidation1	-8.114e-03	3.786e-03	3.735e+04
-2.143			
## int_rate	1.526e-02	2.060e-03	3.675e+04
7.409			
## revol_util_scaled	-6.309e-04	3.421e-04	3.732e+04
-1.844			
## credit_age_yrs	-6.236e-04	1.219e-03	3.735e+04
-0.512			
## annual_inc_10k	6.259e-04	1.472e-03	3.729e+04
0.425			
## dti	1.798e-03	1.363e-03	3.731e+04
1.319			
## loan_amnt_10k	-1.007e-02	1.315e-02	3.733e+04
-0.766			
## close_acc	1.709e-04	1.092e-03	3.736e+04
0.156			
## open_acc	-1.672e-03	2.064e-03	3.735e+04
-0.810			
## int_rate:revol_util_scaled	1.440e-05	1.990e-05	3.735e+04
0.724			
## int_rate:credit_age_yrs	-2.430e-06	9.493e-05	3.730e+04
-0.026			
## int_rate:annual_inc_10k	-2.153e-04	1.203e-04	3.726e+04
-1.790			
## int_rate:dti	4.106e-05	9.711e-05	3.734e+04
0.423			
## int_rate:loan_amnt_10k	2.718e-03	7.736e-04	3.731e+04
3.514			
## int_rate:close_acc	-2.857e-04	7.497e-05	3.729e+04
-3.810			
## int_rate:open_acc	-6.582e-05	1.321e-04	3.731e+04
-0.498			
## revol_util_scaled:credit_age_yrs	3.339e-05	1.242e-05	3.732e+04
2.689			
## revol_util_scaled:annual_inc_10k	9.577e-06	1.448e-05	3.730e+04

0.661			
## revol_util_scaled:dti	-1.465e-05	1.140e-05	3.730e+04
-1.285			
## revol_util_scaled:loan_amnt_10k	-2.711e-04	1.146e-04	3.735e+04
-2.366			
## revol_util_scaled:close_acc	1.039e-05	9.915e-06	3.731e+04
1.048			
## revol_util_scaled:open_acc	7.842e-05	1.858e-05	3.731e+04
4.221			
## credit_age_yrs:annual_inc_10k	4.713e-08	4.904e-05	3.728e+04
0.001			
## credit_age_yrs:dti	4.610e-05	5.027e-05	3.731e+04
0.917			
## credit_age_yrs:loan_amnt_10k	-8.451e-04	4.302e-04	3.731e+04
-1.964			
## credit_age_yrs:close_acc	-3.925e-05	3.646e-05	3.736e+04
-1.077			
## credit_age_yrs:open_acc	5.126e-05	7.185e-05	3.732e+04
0.713			
## annual_inc_10k:dti	-3.894e-04	5.775e-05	3.710e+04
-6.744			
## annual_inc_10k:loan_amnt_10k	2.478e-04	4.023e-04	3.733e+04
0.616			
## annual_inc_10k:close_acc	2.030e-05	4.146e-05	3.733e+04
0.490			
## annual_inc_10k:open_acc	3.443e-05	7.788e-05	3.730e+04
0.442			
## dti:loan_amnt_10k	-4.948e-05	4.696e-04	3.735e+04
-0.105			
## dti:close_acc	1.086e-04	3.936e-05	3.733e+04
2.759			
## dti:open_acc	-1.224e-04	7.036e-05	3.734e+04
-1.740			
## loan_amnt_10k:close_acc	6.199e-04	3.393e-04	3.731e+04
1.827			
## loan_amnt_10k:open_acc	-5.990e-04	6.868e-04	3.731e+04
-0.872			
## close_acc:open_acc	5.736e-05	5.097e-05	3.733e+04
1.125			
##	Pr(> t)		
## (Intercept)	0.122668		
## home_ownershipOTHER	0.023846	*	
## home_ownershipOWN	0.313075		
## home_ownershipRENT	0.024074	*	
## inq_last_6mths	< 2e-16	***	
## term60 months	< 2e-16	***	
## delinq_2yrs	0.923214		
## verification_statusSource Verified	0.773354		
## verification_statusVerified	0.917772		
## emp_length_101	0.000143	***	

```

## purpose_debt_consolidation1      0.032098 *
## int_rate                        1.30e-13 ***
## revol_util_scaled                0.065159 .
## credit_age_yrs                  0.608992
## annual_inc_10k                  0.670682
## dti                             0.187336
## loan_amnt_10k                   0.443822
## close_acc                       0.875652
## open_acc                        0.417860
## int_rate:revol_util_scaled       0.469364
## int_rate:credit_age_yrs         0.979578
## int_rate:annual_inc_10k         0.073398 .
## int_rate:dti                   0.672426
## int_rate:loan_amnt_10k          0.000442 ***
## int_rate:close_acc              0.000139 ***
## int_rate:open_acc              0.618246
## revol_util_scaled:credit_age_yrs 0.007168 **
## revol_util_scaled:annual_inc_10k 0.508477
## revol_util_scaled:dti           0.198724
## revol_util_scaled:loan_amnt_10k 0.017984 *
## revol_util_scaled:close_acc      0.294759
## revol_util_scaled:open_acc       2.44e-05 ***
## credit_age_yrs:annual_inc_10k    0.999233
## credit_age_yrs:dti              0.359065
## credit_age_yrs:loan_amnt_10k     0.049511 *
## credit_age_yrs:close_acc         0.281704
## credit_age_yrs:open_acc          0.475648
## annual_inc_10k:dti              1.56e-11 ***
## annual_inc_10k:loan_amnt_10k     0.537990
## annual_inc_10k:close_acc         0.624460
## annual_inc_10k:open_acc          0.658430
## dti:loan_amnt_10k               0.916098
## dti:close_acc                   0.005796 **
## dti:open_acc                    0.081838 .
## loan_amnt_10k:close_acc          0.067697 .
## loan_amnt_10k:open_acc           0.383156
## close_acc:open_acc              0.260433
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## fit warnings:
## Some predictor variables are on very different scales: consider rescaling

```

Appendix B

```
summary(state_time_model_2)
```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: formula <- repay_fail ~ home_ownership + inq_last_6mths + term +

```

```

##      delinq_2yrs + verification_status + emp_length_10 +
purpose_debt_consolidation +
##      (int_rate + credit_age_yrs + dti + close_acc + open_acc)^2 +
##      (1 | zip_code) + (1 | issue_d)
##      Data: extended_data
##
## REML criterion at convergence: 27160.4
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -2.3178 -0.5237 -0.3328 -0.1209  3.0043
##
## Random effects:
##      Groups   Name      Variance Std.Dev.
## zip_code (Intercept) 0.0008227 0.02868
## issue_d   (Intercept) 0.0009670 0.03110
## Residual                0.1190680 0.34506
## Number of obs: 37425, groups: zip_code, 830; issue_d, 55
##
## Fixed effects:
##
##              Estimate Std. Error      df t
value
## (Intercept)          -6.701e-02  2.368e-02  1.094e+04
-2.830
## home_ownershipOTHER          7.752e-02  3.161e-02  3.669e+04
2.453
## home_ownershipOWN           1.248e-02  7.225e-03  3.733e+04
1.727
## home_ownershipRENT          1.597e-02  4.235e-03  2.667e+04
3.772
## inq_last_6mths           1.925e-02  1.234e-03  3.138e+04
15.608
## term60 months            6.362e-02  5.072e-03  1.908e+04
12.544
## delinq_2yrs             -1.929e-03  3.646e-03  3.736e+04
-0.529
## verification_statusSource Verified -4.218e-03  4.850e-03  1.844e+04
-0.870
## verification_statusVerified    -6.674e-03  4.589e-03  2.487e+04
-1.454
## emp_length_101           1.621e-02  4.569e-03  3.735e+04
3.547
## purpose_debt_consolidation1    -5.567e-03  3.689e-03  3.738e+04
-1.509
## int_rate                1.611e-02  1.588e-03  3.719e+04
10.150
## credit_age_yrs           -3.809e-04  1.186e-03  3.737e+04
-0.321
## dti                     -4.240e-04  1.251e-03  3.735e+04
-0.339

```

## close_acc	6.186e-04	1.061e-03	3.737e+04
0.583			
## open_acc	-6.162e-04	1.996e-03	3.738e+04
-0.309			
## int_rate:credit_age_yrs	3.554e-05	7.944e-05	3.732e+04
0.447			
## int_rate:dti	2.071e-05	7.704e-05	3.734e+04
0.269			
## int_rate:close_acc	-2.125e-04	6.151e-05	3.732e+04
-3.455			
## int_rate:open_acc	5.990e-05	1.152e-04	3.732e+04
0.520			
## credit_age_yrs:dti	6.685e-05	4.531e-05	3.733e+04
1.475			
## credit_age_yrs:close_acc	-3.695e-05	3.425e-05	3.738e+04
-1.079			
## credit_age_yrs:open_acc	-3.281e-05	6.796e-05	3.734e+04
-0.483			
## dti:close_acc	9.459e-05	3.561e-05	3.734e+04
2.656			
## dti:open_acc	-8.662e-05	6.670e-05	3.735e+04
-1.299			
## close_acc:open_acc	3.752e-05	4.744e-05	3.735e+04
0.791			
##	Pr(> t)		
## (Intercept)	0.004666	**	
## home_ownershipOTHER	0.014189	*	
## home_ownershipOWN	0.084120	.	
## home_ownershipRENT	0.000162	***	
## inq_last_6mths	< 2e-16	***	
## term60 months	< 2e-16	***	
## delinq_2yrs	0.596800		
## verification_statusSource Verified	0.384482		
## verification_statusVerified	0.145857		
## emp_length_101	0.000390	***	
## purpose_debt_consolidation1	0.131264		
## int_rate	< 2e-16	***	
## credit_age_yrs	0.748156		
## dti	0.734731		
## close_acc	0.559970		
## open_acc	0.757537		
## int_rate:credit_age_yrs	0.654556		
## int_rate:dti	0.788052		
## int_rate:close_acc	0.000551	***	
## int_rate:open_acc	0.603178		
## credit_age_yrs:dti	0.140114		
## credit_age_yrs:close_acc	0.280610		
## credit_age_yrs:open_acc	0.629230		
## dti:close_acc	0.007900	**	
## dti:open_acc	0.194075		

```
## close_acc:open_acc          0.429000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 26 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it.
```

Appendix C

Numeric Variables Pair Plot

